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COLLEGE OF ENGINEERING

DEPARTMENT OF COMPUTER ENGINEERING

EE 288: ELECTRICAL MEASUREMENT AND INSTRUMENTATION

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ABSTRACT

This report details the finding from Lab Session 1 of the Electrical Measurement and Instrumentation course. This focuses on the introduction to sensors and Analog-to-Digital Conversion(ADC) using the ESP32 microcontroller. The primary objective was to bridge theoretical knowledge of sensor technologies with practical data acquisition techniques. Preliminary research was conducted on various sensor modules, that is the Light Dependent Resistors for ambient light, DHT modules for humidity and temperature, ultrasonic sensor for distance measurement, gas sensors and motion sensors. The lab activity further progress to programming the microcontroller to acquire environmental data via these sensors. The ESP32's internal 12-bit ADC was utilized to get digital values between 0 and 4095. The results successfully demonstrated the real-time acquisition and interpretation of both analog and digital sensor data, verifying the fundamental principles of environmental monitoring and microcontroller integration.

1 Lab Session 1 (Theoretical Background)

This lab session focused on understanding the underlying principle that governs the operation of the following sensors.

1.1 Overview of Sensors

1. **Light Sensors (LDR):** This sensor measures the ambient light intensity and converts it to electrical signals.

- **How it works:** Light exposures frees electrons in the semiconductor material this changes the current flows which are detected as electrical signals.
- **Applications:** Automatic street lighting, factory sorting, automatic screen brightness, greenhouse monitoring.

2. **Humidity sensor:** This sensor tracks water vapor by measuring the relative humidity of air.

- **How it works:** The mode of operation of this sensor depends on it's category.
- **Categories:**
 - **Capacitive humidity sensors:** These type of humidity sensor utilizes a moisture-absorbing polymer film to alter the dielectric constant and capacitance. These types are found in the DHT22 sensor module.
 - **Resistive humidity sensors:** These type utilizes a conductive track with absorbing salts. When the water vapor in the atmosphere increases the salts are dissolved and this tends to decrease the resistance of the track which affects the voltage values.
- **Applications:** Meteorological forecasting, anti-fogging systems, and general environment tracking.

3. **Temperature sensor:** This sensor measures the thermal energy and detects ambient temperature changes.

- **Categories:**
 - **Thermo-resistive types:** Utilizes the change in resistance when there is a temperature change. RTDs and Thermistors(e.g. DHT11 module).
 - **Thermo-electric types:** Generates tiny voltages via the Seebeck's effect. E.g. Thermocouples

- **Semiconductor (IC) types:** Utilizes predictable voltage drops. E.g. LM35 (analog), DS18B20 (digital)
 - **Applications:** HVAC systems, manufacturing industries, general environmental tracking (DHT sensor module), automotive and aviation.
4. **Motion sensor:** This sensor detects physical movement, presence, or changes in position.
- **How it works:** The fundamental operating principle depends on the type of sensor.
 - **Passive Infrared (PIR) Sensors:** They don't emit energy but rather measure infrared radiation (heat) naturally emitted by objects. When a warm body (like a human or pet) walks into the field of view, the sensor detects a rapid fluctuation in heat and triggers an alert.
 - **Microwave Sensors:** These emit microwave pulses and measure how the waves bounce back to the sensor. Movement changes the frequency or timing of the returning waves (Doppler effect). They can cover larger areas than PIR sensors and can even pass through walls.
 - **Dual-Technology Sensors:** These combine multiple sensors (like PIR and microwave) to reduce false alarms. An alarm will only trigger if both technologies detect movement simultaneously.
 - **Ultrasonic Sensors:** These operate like sonar by sending out high-frequency sound waves that are inaudible to human ears. If a moving object disrupts the sound waves, the altered echo time triggers an action. They are highly sensitive to subtle movements.
 - **Applications:** Security and alarm systems, automatic indoor/ outdoor lighting, industrial machines safety triggers.
5. **Gas sensors:** These sensors detect gas presence and measure the concentration of these gases for safety and monitoring.
- **Categories:**
 - **MOS type:** Detects gases via changes in electrical resistance
 - **Electrochemical type:** Produces an electrical current when reacting with target gases
 - **Infrared type:** Measures how much infrared light a specific gas absorbs.
 - **Applications:** Residential and industrial leak detection, air pollution monitoring, vehicle emission control, medical equipments.
6. **Ultrasonic sensors:** This sensor emits high-frequency sound waves(beyond human hearing) and detects reflected echoes to measure distance and detect objects.

- **How it works:** The transmitter sends a burst of ultrasonic sound. The receiver picks up the returning echo. Distance is calculated using the time delay and the speed of sound.
- **Applications:** Obstacle detection in robotics, automotive parking assist, presence detection in manufacturing, tank level measurement in processing industries.

2 Lab session 2(Sensor interfacing)

2.1 Objective

The main objective of this lab activity was to program an ESP32 microcontroller to receive data values from a DHT11 sensor module, LDR sensor and an ultrasonic sensor.

2.2 Analog-to-Digital Conversion (ADC) on ESP32

Analog-to-Digital conversion is the process of converting continuous changing physical signals (analog) into discrete number, that the computer can process, store and transmit. This process is done by an Analog-to-Digital converter(ADC).

The ESP32 features multiple 12-bit ADC channels that read voltage values from 0 to 3.3V and maps them to digital values from 0 to 4095.

2.3 Materials and Setup

The following components were utilized:

- ESP32 Development Board
- DHT11 Temperature and Humidity Sensor
- LDR
- 10k Ω Resistor
- HC-SR04 Ultrasonic Sensor

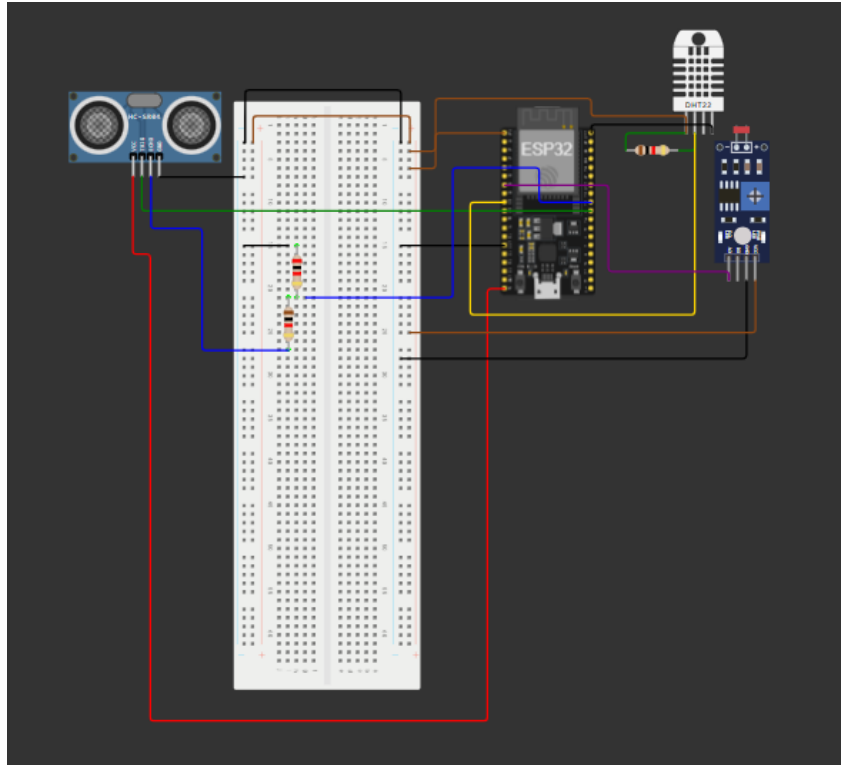


Figure 2.1: Circuit diagram of the ESP32 interfaced with DHT11, LDR, and Ultrasonic sensors.

NB: The circuit diagram uses a LDR sensor module with an internal voltage divider

```

1  #include <DHT.h>
2
3  #define DHTTYPE DHT22
4
5  const int dht_pin = 25;
6  const int trig_pin = 5;
7  const int echo_pin = 18;
8  const int ldr_pin = 32;
9
10 // constant variable to hold the speed of sound in cm per microsecond
11 const float speed_of_sound = 0.0343;
12
13 // Declaring dht object
14 DHT dht(dht_pin, DHTTYPE);
15
16 // Variable declaration to hold sensor values.
17 long duration;
18 float distance_cm;
19 float temperature, humidity;
20 int ldr_value = 0;
21
22 void setup()
23 {
24     // setup up the display on the serial monitor
25     Serial.begin(115200);
26
27     // function call of the begin() method to initialize the DHT
28     dht.begin();
29
30     // pinMode setup the vaious GPIO pin allocations.
31     pinMode(trig_pin, OUTPUT);
32     pinMode(echo_pin, INPUT);
33     pinMode(ldr_pin, INPUT);
34 }
35
36 void loop()
37 {
38     digitalWrite(trig_pin, LOW);
39     delayMicroseconds(2);
40     // 2. TRIGGER THE SENSOR (The sequence of High, Low, sets the HC-SR04 to send a blast of ultrasonic waves)
41     // A 10-microsecond HIGH pulse is the standard required to trigger the HC-SR04
42     digitalWrite(trig_pin, HIGH);
43     delayMicroseconds(10);
44     // sending low signal
45     digitalWrite(trig_pin, LOW);
46
47     // 3. LISTEN FOR THE ECHO
48     // pulseIn() waits for the echo pin to go HIGH, times how long it stays HIGH,
49     // and returns that time in microseconds.
50     duration = pulseIn(echo_pin, HIGH);
51
52     // 4. CALCULATE DISTANCE
53     // We divide by 2 because the sound traveled OUT and BACK (twice the distance).
54     distance_cm = (duration * speed_of_sound) / 2;
55
56     // DHTpin- getting humidity and temp values
57     temperature = dht.readTemperature();
58     humidity = dht.readHumidity();
59
60     // Error checking. This is to check if ESP32 is getting reading from the microcontroller
61     if (isnan(humidity) || isnan(temperature))
62     {
63         Serial.println("Error reading from DHT sensor");
64         return;
65     }
66
67     // Getting analog reading from the LDR sensor
68     ldr_value = analogRead(ldr_pin);
69
70     // 5. PRINT THE RESULTS
71     Serial.print("Distance: ");
72     Serial.print(distance_cm);
73     Serial.println(" cm");
74
75     Serial.print("Temp: ");
76     Serial.print(temperature);
77     Serial.print("C | Humidity: ");
78     Serial.print(humidity);
79     Serial.println("%");
80     Serial.print("Light level: ");
81     Serial.println(ldr_value);
82
83     if (ldr_value < 300)
84         Serial.println("Its dark");
85     else
86         Serial.println("Its bright");
87     Serial.println("-----");
88
89     // A delay of 2 seconds before we take the next set of readings from the various sensors.
90     delay(2000);
91 }

```

Figure 2.2: Code sample

2.4 Results

The data acquired from the sensors was printed to the Serial Monitor.

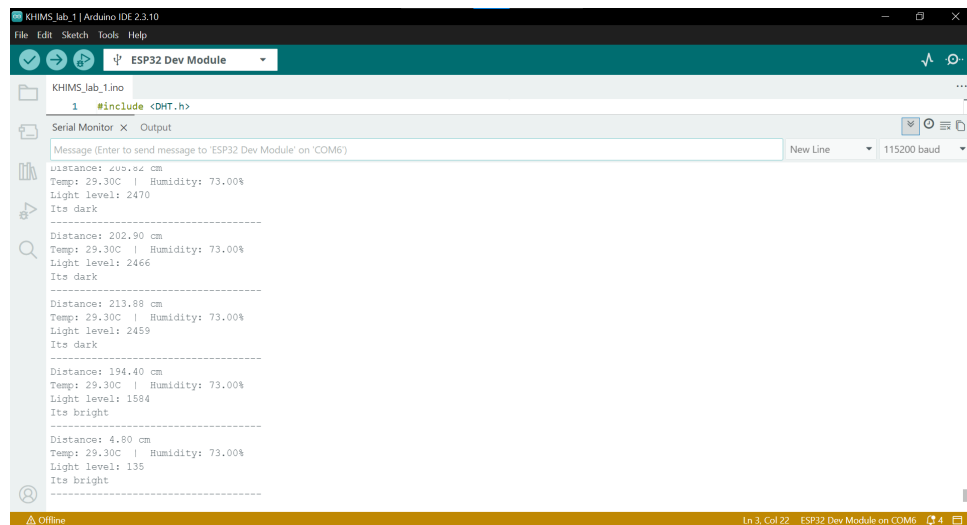


Figure 2.3: Serial monitor output displaying temperature, humidity, light, and distance readings.

2.5 Discussion

The sensors responded predictably to environmental changes.

- **Ultrasonic sensor:** As the obstacle moved closer to the sensor, the distance decreased and as the obstacle moved further away from the sensor, the distance increased.
- **DHT11 sensor:** As hot air was blown over the sensor, the humidity values decreased and the temperature value increased.
- **LDR sensor:** As the environment was darkened, the digital values increased to 4095. Similarly, when the environment was lightened up, the digital value on the serial monitor decreased.

2.6 Conclusion

This lab successfully demonstrated the integration of both analog and digital sensors with the ESP32.

You can access the GitHub repository using the link below.

https://github.com/sadzoiedem/team_khims_project